Day 47

SOAP Adapter

* If we want to call any Soap Based Web Service's or Soap Based API then we will use Soap Adapter

* API Based Integration's → Realtime integration's
  - SOAP
  - REST
  - ODATA

* NOW we discuss about SOAP Adapter

⇒ If we want to make any Web service Based Communication or SOAP Based Communication then we will use SOAP Adapter ✓

* When we don't have API at that time they used to call as Web Service.

* Any Web service Based Communication we will go with SOAP Adapter ✓

* For SOAP Adapter the prerequisite is WSDL File ✓

* Let's try to understand with WSDL File.

⇒ WSDL → Web Service definition language ✓

* It is nothing but a Schema of a Service and what all are operations we are calling and what is end point

* Schema → XSD → WSDL file ✓

* In Detail about WSDL file ✓

* Eg: calculator WSDL File

↳ Add, multiply, substract, etc

These all operations are present in WSDL file

* Let's take that WSDL file from online

* Calculator web service → dneonline.com

here we can find that file you can view and download it to the system ✓

* Save it as .wsdl ✓

* What is the need of WSDl File

* Every time if you open WSdl file then you can observe that the file data will be started with <wsdl: definitions

⇒ <wsdl: ~~schema~~ types>

<s: schema - - -

</s: schema> } → XSD ✓

* XSD → It's nothing But schema ✓

* XSD always start with <s:schema - - -

</s:schema>

* <wsdl:Services - - - -

</wsdl: Services>

→ Structure: <wsdl: Services → - - - -

<wsdl:port① - - - - name=.., binding=..

<soap:adress location - - -

</wsdl:port>

<wsdl:port② - - - - name=.., binding=..

→ same

</wsdl: port②- - - -

</wsdl: service>

*port → which port your connecting ✓

Port → location ⇒ loction is to which endpoint you are connecting ✓

* Soap. binding: It is used to bind the operations for that port ✓

* If we want to call Target System → then we will get WSDL → file from them ✓

Service, port, binding, operations ✓

* port → operations → bindings → message → Element
Input output

all are interlinked

* Service → port → binding → operations → message →
element name → field name (Structure) Input output

* xml → every field we will called as Element

* Calling → webservice (api) → Structure → Fix Structure.

* REST → Don't have Structure.

* SOAP → defined format → only Accepts XML
Content

⇒ In message If we have Input & output then it is Synchronous. Other wise If we have only Input then it is Asynchronous
Input (Request) & (output) Response

* URL & Structure in WSDL

* SOAP Webservice → Thirdpart tool
↓
SoapUI → free open Source tool

* SOAP Based api's & Regression Testing

* demonstration of an example in Soapui Tool

* ① create a new project
  ② Upload wsdl file

* The port name's : Bindings

① Calculator Soap &
② Calculator Soap12

It have Operation's
add
divide
multiply
Substract

It have Operation's

Same operations

opening the file you will get Structure which is fixed

* When ever we are using Soap based file wsdl Remember, Structure is same, so NOW we need to send the data by Including below mandatory Tags

① Soap Envelop ✓
② Soap Header ✓
③ Soap Body ✓
↓
xml payload ✓

If we want to send any soap data we need to send with beside Mandatory tags otherwise interface will fail ✓

* Mandatory Envelop tag ✓

⇒ SOAP mainly for legacy systems

* Import → Operations → payload → URL ✓
WSDL file

* Now the task is from SAP CPI how we can Call SOAP "webservice"

* Demo Explained ✓ in sap cpi ✓

* API → Endpoint URL → Authentication } Connection Tab
* SOAP → Endpoint URL → Authentication → WSDL

⇒ First Explaining Reciever SOAP Adapter ✓

① Address → End point URL
② proxy Type → Internet or on promise.
③ URL to WSDL : Upload (Calculator.WSDL)
④ Service:

(5) Endpoint : Binding

(6) operation name:

(7) Authentication : None (legacy System) ✓
certificate
Saml/Bearer ✓

(8) Timeout: 6000

(9) Keep-Alive: ☑

(10) Compress message: ☑

(11) Allow chunking: ☐

(12) Return HTTP Response Code: ☑
as Header

⇒ WS-security (let's use "none")

* Demo going on ✓

Addition operation performed ✓

* No need to append the mandatory tags
So envelop, Headers, body !
soap cp → will → add those ✓
Add & Remove will be be done
by the Soap Adapter ✓

* WS Security: It Suppose When we are working

SOAP Web ~~base~~ Service → In the legacy Applications

The WSDL to have Security headers

↓

⇒ If I want to send WSDL Security headers

We can use SOAP Adapter

↓

→ Based on policies in WSDL

→ None

→ via manual Configuration in channel

↓

options: → Hashed password

→ None

→ Plain Text password

↳ few minutes Dynamic one only

⇒ Signing an Encryption

→ None ✓

→ Sign and Encrypt message ✓

→ Sign message ✓

* If We got requirement saying that We are using Security headers to WSDL file the We need to Configure WS-Security

↳ Secure

* NOW SOAP Sender Adapter

It's Same like HTTP

* Connection Tab

① Address

Service Definition

Use WS-Addressing

Message Exchange pattern: Request-Reply
Request
Response
two way
one way

Synchronous

Asynchronous based Connection

Processing Setting

Robust

WS Standard

Will not send error back Sender Sys.

Error Content will be Send to the Sender System

* Endpoint URL
— WSDL
— WSDL without policies